The Universal Deviation Law: Geometric, Information-Theoretic, and Variational Guarantees for Anomaly Detection

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Abstract

We introduce the **Universal Deviation Law (UDL)**, a theoretically grounded framework for domain-agnostic anomaly detection. The core principle is that a point which is *meaningfully different from the normal set* cannot simultaneously match normality under all structural perspectives. We formalise this via *deviation-separating operator families*, collections of differentiable maps $\{\Phi_k\}_{k=1}^K$ whose stacked Jacobian has full column rank, yielding local distinguishability around a reference normal state $\boldsymbol{\mu}$. To connect the sensitivity theory to detection, we define a normal set $\mathcal{S}$ and a deviation score based on distance of $\Phi(\mathbf{x})$ to $\Phi(\mathcal{S})$; under an explicit separation condition between normal and anomaly sets in operator space, at least one operator deviates by a positive margin. We prove five results: (1) local distinguishability via the Inverse Function Theorem; (2) global sensitivity via compactness and the Extreme Value Theorem with a uniform bound $\underline{\sigma} > 0$; (3) finite-sample guarantees via Hoeffding concentration; (4) information-optimal operator fusion weights via a semidefinite programme; and (5) a greedy operator selection algorithm with $(1 - 1/e)$-approximation guarantee via monotone submodularity. We instantiate the framework with five spectrum projection operators (statistical, dynamical, spectral, geometric, exponential) and introduce a **Boundary-Centred Dimension Magnifier** paired with QDA that, given labels, achieves strong supervised performance. The unsupervised MDN score provides a label-free alternative (mean AUC 0.884 across four benchmarks).

We also present the **GravityEngine**, an N-body variational scoring algorithm whose force potentials we derive in closed form. The attraction potential involves the error function $\mathrm{erf}(r/\sigma)$. Backtracking line-search on the resulting energy functional guarantees monotone descent at every iteration, which we prove by establishing the energy as a Lyapunov function. Empirically, UDL-QDA-Magnified achieves mean AUC 0.984 on four benchmarks (two synthetic datasets plus two ODDS datasets) under a single hyperparameter set. GravityEngine reaches AUC 0.897 on mammography with verified monotone convergence (14/14 iteration steps, energy strictly decreasing). We numerically verify the theorem conditions on three independent ODDS benchmarks via Monte Carlo checks and observe no violations in our samples. Code is available at https://github.com/segetii/fintech.

1 Introduction

Every anomaly detector makes assumptions about what anomalies look like. Isolation Forest [Liu et al., 2008] assumes geometric isolation. LOF [Breunig et al., 2000] assumes local density deviation. Deep SVDD [Ruff et al., 2018] assumes separation from a learned hypersphere. Each works well when its assumptions hold, and each has blind spots when they do not. An anomaly that mimics normal local density will evade LOF entirely, yet the same anomaly may be obvious under spectral analysis or geometric projection. This is not a minor inconvenience; it is a structural limitation that recurs across fraud detection [Ahmed et al., 2016], medical diagnosis [Fernando et al., 2021], cybersecurity [Chandola et al., 2009], industrial monitoring [Pang et al., 2021], and aerospace telemetry [Hundman et al., 2018].

A natural question follows: can we state conditions under which combining multiple mathematical perspectives makes anomalies detectable? This paper answers in the affirmative under explicit sensitivity and separation assumptions. We introduce the **Universal Deviation Law**, which provides local, global, finite-sample, and algorithmic guarantees for operator-family sensitivity, and a set-based separation condition that turns sensitivity into a detection statement.

Contributions. Our contributions are:

1. **Theoretical framework** (Section 3): Five results establishing provable *sensitivity/distinguishability* for deviation-separating operator families, plus an explicit set-based separation condition that yields a detection guarantee. This includes local (Theorem 6) and global (Theorem 7) sensitivity bounds, finite-sample bounds (Theorem 8), optimal fusion via SDP (Theorem 10), and greedy selection with $(1 - 1/e)$-approximation (Theorem 11). The individual proof techniques (IFT, EVT, Hoeffding, SDP relaxation, submodular optimisation) are classical; the contribution is the *framework* that threads them into end-to-end conditions for detectability.
2. **GravityEngine** (Section 4): An N-body variational scoring algorithm with a *provably correct* energy functional (attraction via erf potential, logarithmic repulsion, quadratic confinement). We prove monotone energy descent under backtracking line-search, establishing the energy as a Lyapunov function (Theorem 12).
3. **Multi-spectrum architecture** (Section 5): Five composable spectrum operators, a Boundary-Centred Dimension Magnifier with saturating $\tanh$ non-linearity, and QDA-Magnified projection achieving state-of-the-art AUC.
4. **Empirical evaluation** (Section 6): Five datasets across four domains, a single hyperparameter set throughout, comparisons against seven baselines including Deep SVDD and ECOD, 5-fold cross-validation on mammography, per-law decomposition analysis (including pendigits), and numerical checks of theorem conditions on three ODDS datasets.

2 Related Work

Classical methods. Distance-based approaches (k-NN [Ramaswamy et al., 2000], Mahalanobis) and density-based methods (LOF [Breunig et al., 2000], DBSCAN [Ester et al., 1996]) operate in geometric space. Tree-based methods like Isolation Forest [Liu et al., 2008] exploit random partitioning. All of these impose a single mathematical viewpoint on the data.

Deep approaches. Autoencoder-based detectors [An and Cho, 2015, Zhou and Paffenroth, 2017] flag high reconstruction error. Deep SVDD [Ruff et al., 2018] learns a minimum-volume hypersphere. GANs [Schlegl et al., 2017] model the normal distribution and flag deviations from it. These capture complex patterns, but they still produce a single representation of normality.

Modern unsupervised detectors. ECOD [Li et al., 2022] uses empirical cumulative distribution functions per feature. COPOD [Li et al., 2020] extends this with copula-based dependence modelling. Both achieve competitive performance in $O(n)$ time. Neither provides theoretical detectability guarantees.

OOD detection. Hendrycks and Gimpel [Hendrycks and Gimpel, 2017] showed that softmax probabilities give a usable OOD baseline; ODIN [Liang et al., 2018] and energy-based methods [Liu et al., 2020] improved on it. These produce a single OOD score. UDL differs because it decomposes *which mathematical aspect* makes an observation anomalous.

Ensemble and multi-view methods. Feature bagging [Lazarevic and Kumar, 2005] and detector aggregation [Aggarwal, 2015] combine multiple instances of similar detectors. Krogh and Vedelsby [Krogh and Vedelsby, 1995] proved that ensemble error decreases with member diversity. UDL's diversity operates at the *representation* level rather than the *model* level: it fuses mathematically distinct projection operators, not multiple copies of the same detector.

Physics-inspired clustering. Particle swarm optimisation [Kennedy and Eberhart, 1995], gravitational search [Rashedi et al., 2009], and charged system search [Kaveh and Talatahari, 2010] use physical analogies for optimisation. GravityEngine differs in three ways: its potential functions are derived from the force field in closed form, force-energy consistency ($F = -\nabla E$) is verified analytically, and monotone energy descent is guaranteed via line-search.

Benchmarking. PyOD [Zhao et al., 2019] implements over 30 detectors. ADBench [Han et al., 2022] provides 57 benchmark datasets. We evaluate on ODDS benchmarks [Rayana, 2016] and follow ADBench protocols.

3 Universal Deviation Law: Theory

3.1 Deviation-Separating Operator Families

Definition 1 (Deviation-Separating Family)**.** *Let $\{\Phi_k\}_{k=1}^K$, $\Phi_k : \mathbb{R}^d \to \mathbb{R}^{d_k}$, be continuously differentiable operators. The family is* deviation-separating at $\boldsymbol{\mu} \in \mathbb{R}^d$ *if:*

$$\bigcap_{k=1}^{K} \ker(D\Phi_k(\boldsymbol{\mu})) = \{\mathbf{0}\} \tag{1}$$

Equivalently (Theorem 2), the stacked Jacobian $J(\boldsymbol{\mu}) = [D\Phi_1(\boldsymbol{\mu})^\top, \ldots, D\Phi_K(\boldsymbol{\mu})^\top]^\top \in \mathbb{R}^{D\times d}$ has full column rank, where $D = \sum_k d_k$.

Lemma 2 (Rank Characterisation)**.** *$\{\Phi_k\}$ is deviation-separating at $\boldsymbol{\mu}$ if and only if $\operatorname{rank}(J(\boldsymbol{\mu})) = d$.*

The proof is immediate: $\ker(J(\boldsymbol{\mu})) = \bigcap_k \ker(D\Phi_k(\boldsymbol{\mu}))$.

3.2 From Sensitivity to Detection: A Normal Set $\mathcal{S}$

Theorems 6–8 are sensitivity statements around a reference point $\boldsymbol{\mu}$: they guarantee that *any* perturbation away from $\boldsymbol{\mu}$ produces a response in at least one operator under rank conditions. To interpret this as anomaly detection, we must specify what "normal" means (a set or distribution) and a detection statistic.

Definition 3 (Normal Set and Deviation Score)**.** *Let $\mathcal{S} \subset \mathbb{R}^d$ denote the set of normal states (e.g., the support of a normal distribution $P_{\mathcal{S}}$ or an empirical normal training set). Define the joint operator map $\Phi(\mathbf{x}) := (\Phi_1(\mathbf{x}), \ldots, \Phi_K(\mathbf{x})) \in \mathbb{R}^D$, $D = \sum_k d_k$. Define the set-distance deviation score*

$$s_{\mathcal{S}}(\mathbf{x}) := \operatorname{dist}(\Phi(\mathbf{x}), \Phi(\mathcal{S})) = \inf_{\mathbf{z}\in\mathcal{S}} \|\Phi(\mathbf{x}) - \Phi(\mathbf{z})\|_2 . \tag{2}$$

For per-operator analysis, define $s_{\mathcal{S},k}(\mathbf{x}) := \operatorname{dist}(\Phi_k(\mathbf{x}), \Phi_k(\mathcal{S}))$.

Theorem 4 (Separation in Operator Space Implies Detectability)**.** *Assume $\mathcal{S}$ is compact and each Φ_k is continuous. Let $\mathcal{A} \subset \mathbb{R}^d$ be a compact set of anomalies with $\mathcal{A} \cap \mathcal{S} = \emptyset$, and assume there exists a margin $\Delta > 0$ such that*

$$\operatorname{dist}(\Phi(\mathcal{A}), \Phi(\mathcal{S})) \geq \Delta. \tag{3}$$

Then for all $\mathbf{x} \in \mathcal{A}$, $s_{\mathcal{S}}(\mathbf{x}) \geq \Delta$ and in particular there exists $k \in \{1, \ldots, K\}$ such that

$$s_{\mathcal{S},k}(\mathbf{x}) \geq \Delta/\sqrt{K}. \tag{4}$$

Remark 5 (What the guarantee does and does not claim)**.** *The separation condition (3) is an explicit modelling assumption: if anomalies are not separated from normals in operator space (e.g., an adversary constructs $\mathbf{x} \notin \mathcal{S}$ with $\Phi(\mathbf{x}) \in \Phi(\mathcal{S})$), no operator-based detector can succeed. UDL provides (i) conditions under which perturbations are not invisible (rank conditions), and (ii) a clean separation-to-detection statement once a normal set $\mathcal{S}$ and a detection score are fixed.*

Empirical evidence for operator-space separation. To verify that the separation assumption $\Delta > 0$ holds in practice, we compute Cohen's d (standardised centroid distance) between anomaly and normal embeddings $\Phi(\mathcal{A})$, $\Phi(\mathcal{S})$ on all five benchmarks. The empirical values are: Synthetic $d = 14.6$, Mimic $d = 13.7$, Mammography $d = 3.3$, Shuttle $d = 0.43$, Pendigits $d = 0.81$. All are strictly positive, confirming that the joint operator map induces measurable separation on every dataset. The variation across datasets reflects detection difficulty: large d (synthetic, mimic) corresponds to trivial detection (AUC ≈ 1.0), while smaller d (shuttle, pendigits) correlates with harder tasks. Per-law analysis shows the geometric spectrum contributes the largest separation on four of five datasets, consistent with the per-law AUC results in Table 4.

3.3 Main Theorems

Theorem 6 (Universal Deviation, Local Form)**.** *Let $\{\Phi_k\}_{k=1}^K$ be a deviation-separating family at $\boldsymbol{\mu}$ with each $\Phi_k \in C^1$. Then there exists an open neighbourhood $U \ni \boldsymbol{\mu}$ such that for every $\mathbf{x} \in U \setminus \{\boldsymbol{\mu}\}$:*

$$\exists\, k \in \{1, \dots, K\} : \quad \Phi_k(\mathbf{x}) \neq \Phi_k(\boldsymbol{\mu}) \tag{5}$$

Since $J(\boldsymbol{\mu})$ has full column rank and singular values depend continuously on the matrix, there exists $\varepsilon > 0$ such that $\operatorname{rank}(J(\mathbf{x})) = d$ for all $\mathbf{x} \in B(\boldsymbol{\mu}, \varepsilon)$. This neighbourhood is precisely the U above.

With $\sigma_{\min} := \sigma_{\min}(J(\boldsymbol{\mu})) > 0$, a quantitative bound holds:

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \frac{\sigma_{\min}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| - O(\|\mathbf{x} - \boldsymbol{\mu}\|^2) \tag{6}$$

Theorem 7 (Universal Deviation, Global Form)**.** *Let each $\Phi_k \in C^2$, and let $\mathcal{K} \subset \mathbb{R}^d$ be compact with $\operatorname{rank}(J(\mathbf{x})) = d$ for all $\mathbf{x} \in \mathcal{K}$. Define $\underline{\sigma} := \min_{\mathbf{x}\in\mathcal{K}} \sigma_{\min}(J(\mathbf{x}))$. Then $\underline{\sigma} > 0$, and for all $\mathbf{x} \in \mathcal{K}$:*

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \frac{\underline{\sigma}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| - C \|\mathbf{x} - \boldsymbol{\mu}\|^2 \tag{7}$$

where $C = \frac{1}{2} \sup_{\mathbf{z}\in\mathcal{K}} \left\|D^2\Phi(\mathbf{z})\right\|$ is finite by continuity of $D^2\Phi$ on the compact set.

Theorem 8 (Finite-Sample Deviation Guarantee)**.** *Let $\hat{\boldsymbol{\mu}}_n = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i$ with $\mathbf{x}_i \overset{iid}{\sim} P_{\mathcal{S}}$ sub-Gaussian, population mean $\boldsymbol{\mu}$, covariance Σ. Let each Φ_k be L_k-Lipschitz, $L = \max_k L_k$. Then with probability $\geq 1 - \delta$:*

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\hat{\boldsymbol{\mu}}_n)\| \geq \frac{\underline{\sigma}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| - \frac{L\sqrt{\operatorname{tr}(\Sigma)}\sqrt{2\ln(2d/\delta)}}{\sqrt{n}} \tag{8}$$

Corollary 9 (Sample Complexity)**.** *The bound (8) is non-trivial when $n > 2L^2K \cdot \operatorname{tr}(\Sigma) \cdot \ln(2d/\delta)/(\underline{\sigma}^2 \|\mathbf{x} - \boldsymbol{\mu}\|^2)$.*

Theorem 10 (Information-Optimal Fusion Weights)**.** *Define per-operator information matrices $\mathcal{I}_k := D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu}) \succeq 0$. The worst-case detection sensitivity under weighted fusion $S_{\mathbf{w}}[\mathbf{x}] = \sum_k w_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|^2$ is:*

$$\eta(\mathbf{w}) := \lambda_{\min}(\textstyle\sum_k w_k \mathcal{I}_k) \tag{9}$$

which is concave on the simplex Δ_K (since $\lambda_{\min}$ is an infimum of linear functions). Geometrically, $\eta(\mathbf{w})$ is the worst-case directional sensitivity: the minimum squared response of the fused operator to any unit perturbation. Maximising η over $\mathbf{w} \in \Delta_K$ yields the weight vector $\mathbf{w}^*$ that makes the weakest detection direction as strong as possible, solvable as a standard SDP.

Theorem 11 (Greedy Operator Selection)**.** *Given M candidate operators with $\mathcal{I}_k \succeq 0$ for each k and budget K, selecting S^* to maximise $\lambda_{\min}(\sum_{k\in S} \mathcal{I}_k)$ is NP-hard. The surrogate $f(S) = \log\det(\sum_{k\in S} \mathcal{I}_k)$ is monotone submodular (monotonicity and diminishing returns both require $\mathcal{I}_k \succeq 0$), and greedy selection achieves:*

$$f(S_{\textit{greedy}}) \geq (1 - 1/e) \cdot f(S^*) \tag{10}$$

Surrogate gap. The $(1 - 1/e)$ guarantee holds for $f(S) = \log\det(\cdot)$, not directly for $\lambda_{\min}(\cdot)$. In practice the two objectives are correlated: maximising the log-determinant tends to raise the smallest eigenvalue because $\log\det = \sum \log \lambda_i$ penalises small eigenvalues logarithmically. Bounding the gap between the two objectives in the worst case remains open; we report both SDP-optimal and greedy performance empirically (Table 6).

All proofs are in Section A.

3.4 How $J(\mathbf{x})$ is computed in practice

For general (nonlinear) operators where analytic derivatives are inconvenient, we estimate each Jacobian $D\Phi_k(\boldsymbol{\mu})$ by central finite differences. Concretely, for input dimension d and reference point $\boldsymbol{\mu}$, we perturb coordinate j by

$\varepsilon_j = \varepsilon(|\mu_j| + 10^{-6})$ and compute

$$\frac{\partial \Phi_k}{\partial x_j}(\boldsymbol{\mu}) \approx \frac{\Phi_k(\boldsymbol{\mu} + \varepsilon_j \mathbf{e}_j) - \Phi_k(\boldsymbol{\mu} - \varepsilon_j \mathbf{e}_j)}{2\varepsilon_j}. \tag{11}$$

We then stack the per-operator Jacobians vertically to form $J(\boldsymbol{\mu})$ and compute $\sigma_{\min}(J)$ via SVD. This matches the implementation used in our operator-diversity diagnostics (see the finite-difference Jacobian routine in `udl/energy.py`).

4 GravityEngine: N-Body Variational Scoring

The theoretical framework of Section 3 provides sensitivity/distinguishability guarantees and, under the explicit separation condition of Theorem 4, a detectability statement. We now turn to the question of *how to score* observations. The GravityEngine implements the deviation principle through physics-inspired N-body dynamics with provable convergence.

4.1 Force Field

Each data point $\mathbf{x}_i$ is treated as a particle subject to three forces:

Radial anchoring. A spring-like restoring force toward the data centroid:

$$\mathbf{F}_{\text{radial}}(\mathbf{x}_i) = -\alpha(\mathbf{x}_i - \boldsymbol{\mu}) \tag{12}$$

Pairwise interaction. Short-range attraction with close-range repulsion between particles:

$$\mathbf{F}_{\text{pair}}(\mathbf{x}_i) = -\gamma \sum_{j \neq i} \left[\exp\left(-\frac{r_{ij}^2}{\sigma^2} \right) - \frac{\lambda}{r_{ij}} \right] \hat{\mathbf{r}}_{ij} \tag{13}$$

where $r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ and $\hat{\mathbf{r}}_{ij}$ is the unit vector from j to i.

Deviation penalty. Connects to the UDL framework via the quadratic operator $\Phi(\mathbf{x}) = \mathbf{x}^2$. The deviation energy is $V_{\text{dev}} = \frac{\beta}{2} \left\| \mathbf{x}^2 - \boldsymbol{\mu}^2 \right\|^2$; the force is its negative gradient:

$$F_{\text{dev},j}(\mathbf{x}_i) = -\frac{\partial V_{\text{dev}}}{\partial x_{ij}} = -2\beta\, x_{ij}(x_{ij}^2 - \mu_j^2) \tag{14}$$

i.e. $\mathbf{F}_{\text{dev}} = -2\beta\, \mathbf{x}_i \odot (\mathbf{x}_i^2 - \boldsymbol{\mu}^2)$, where $\odot$ is the Hadamard product.

After iterative evolution, normal points cluster together. Anomalies do not. They end up in isolated positions. The anomaly score is simply the displacement $\left\| \mathbf{x}_i^{(\text{final})} - \mathbf{x}_i^{(\text{initial})} \right\|$.

4.2 Deriving the Correct Energy Functional

For the system to have a Lyapunov function, the energy E must satisfy $\mathbf{F} = -\nabla E$. This is not optional. If the energy and forces are inconsistent, the system is not performing gradient descent on anything, and convergence claims are meaningless. We derive each potential by integration:

Radial potential. From $\mathbf{F} = -\alpha(\mathbf{x} - \boldsymbol{\mu})$:

$$V_{\text{radial}}(\mathbf{x}) = \frac{\alpha}{2} \|\mathbf{x} - \boldsymbol{\mu}\|^2 \tag{15}$$

Pairwise attraction potential. The radial force magnitude from attraction is $f_a(r) = \gamma \exp(-r^2/\sigma^2)$. We need $V_a(r)$ such that $dV_a/dr = f_a(r)$:

$$V_{\text{attract}}(r) = \gamma \int_0^r \exp\left(-\frac{s^2}{\sigma^2} \right) ds = \gamma \cdot \frac{\sigma\sqrt{\pi}}{2} \cdot \operatorname{erf}\left(\frac{r}{\sigma} \right) \tag{16}$$

where $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$ is the error function.

Verification: $\frac{d}{dr}\left[\gamma \frac{\sigma\sqrt{\pi}}{2} \operatorname{erf}(r/\sigma) \right] = \gamma \frac{\sigma\sqrt{\pi}}{2} \cdot \frac{2}{\sigma\sqrt{\pi}} e^{-r^2/\sigma^2} = \gamma e^{-r^2/\sigma^2}$. ✓

Algorithm 1 GravityEngine Iteration with Backtracking Line-Search

Require: Data $\mathbf{X}^0$, learning rate η, damping $\rho \in (0, 1]$, shrink factor $\beta_{\text{ls}} \in (0, 1)$, max halvings L

```
for t = 0, 1, ..., T − 1 do
    F ← −∇E(X^t)                                   ▷ Total force
    η_t ← η                                        ▷ Initial step size
    for ℓ = 0, 1, ..., L − 1 do
        X_trial ← X^t + η_t · ρ · F
        if E(X_trial) ≤ E(X^t) then
            X^{t+1} ← X_trial; break
        else
            η_t ← β_ls · η_t                       ▷ Halve step
        end if
    end for
end for
return X^T
```

Pairwise repulsion potential. From $f_r(r) = -\gamma\lambda/r$:

$$V_{\text{repel}}(r) = -\gamma\lambda \ln r \tag{17}$$

Deviation potential. From $F_j = -2\beta\, x_j(x_j^2 - \mu_j^2)$, we need V with $\partial V/\partial x_j = 2\beta\, x_j(x_j^2 - \mu_j^2)$. Noting $\frac{\partial}{\partial x_j}\left[\frac{1}{2}(x_j^2 - \mu_j^2)^2\right] = 2x_j(x_j^2 - \mu_j^2)$:

$$V_{\text{dev}}(\mathbf{x}) = \frac{\beta}{2}\left\|\mathbf{x}^2 - \boldsymbol{\mu}^2\right\|^2 \tag{18}$$

Verification: $\frac{\partial V_{\text{dev}}}{\partial x_j} = \beta(x_j^2 - \mu_j^2) \cdot 2x_j = 2\beta\, x_j(x_j^2 - \mu_j^2)$. ✓

The total energy is:

$$\boxed{E(\mathbf{X}) = \frac{\alpha}{2}\sum_i \|\mathbf{x}_i - \boldsymbol{\mu}\|^2 + \gamma\frac{\sigma\sqrt{\pi}}{2}\sum_{i<j}\operatorname{erf}\left(\frac{r_{ij}}{\sigma}\right) - \gamma\lambda\sum_{i<j}\ln r_{ij} + \frac{\beta}{2}\sum_i \left\|\mathbf{x}_i^2 - \boldsymbol{\mu}^2\right\|^2} \tag{19}$$

4.3 Monotone Convergence via Backtracking Line-Search

Discrete Euler integration $\mathbf{X}^{t+1} = \mathbf{X}^t + \eta\mathbf{F}(\mathbf{X}^t)$ can overshoot and increase energy when η is too large. A fixed step size would require theoretical bounds on the Lipschitz constant of ∇E, which are hard to obtain for an N-body system. Backtracking line-search (Algorithm 1) avoids this problem entirely by checking the energy at each step and halving η until descent is achieved.

Theorem 12 (Monotone Energy Descent)**.** *Let E be defined by (19) with $\alpha > 0$, and let $\mathbf{X}^{t+1}$ be obtained from $\mathbf{X}^t$ by Algorithm 1. Then:*

$$E(\mathbf{X}^{t+1}) \le E(\mathbf{X}^t) \quad \forall\, t \tag{20}$$

Since E is bounded below (the confining potential $\frac{\alpha}{2}\sum_i \|\mathbf{x}_i - \boldsymbol{\mu}\|^2$ grows without bound), the sequence $\{E(\mathbf{X}^t)\}$ converges.

Proof sketch. The line-search explicitly checks $E(\mathbf{X}_{\text{trial}}) \le E(\mathbf{X}^t)$ before accepting the step. If no step satisfies this condition within L halvings, the minimal step $\eta \cdot \beta_{\text{ls}}^L$ is taken; for sufficiently large L, this step is small enough to satisfy the Armijo-type decrease condition. Boundedness below follows from the quadratic growth of V_{radial} dominating the logarithmic decay of V_{repel} as $\|\mathbf{x}\| \to \infty$. The full proof is in Section B. □

4.4 Connection to UDL

The deviation energy $V_{\text{dev}} = \frac{\beta}{2}\left\|\mathbf{x}^2 - \boldsymbol{\mu}^2\right\|^2$ implements the squared deviation under a single operator $\Phi(\mathbf{x}) = \mathbf{x}^2$. By Theorem 6, if the full operator family including Φ is separating, then any anomaly *within the local neighbourhood U*

of $\boldsymbol{\mu}$ has $V_{\text{dev}} > 0$; for anomalies in a compact region, Theorem 7 extends this globally. The radial potential ensures $V_{\text{radial}} > 0$ for $\mathbf{x} \neq \boldsymbol{\mu}$. Thus the total energy of any anomaly in the relevant compact set is *strictly higher* than the energy at the centroid. We stress that GravityEngine implements only one of the five operators in its energy functional; it is a scoring algorithm inspired by the UDL principle, not a full instantiation of the five-operator framework. The full UDL guarantee requires the multi-spectrum architecture of Section 5.

4.5 Properties of the erf Potential

The erf attraction potential (16) has several desirable properties:

- **Smoothness:** C^∞ everywhere, including at $r = 0$ (unlike Coulomb $1/r$).
- **Boundedness:** $\mathrm{erf}(r/\sigma) \to 1$ as $r \to \infty$; the attraction potential saturates.
- **Built-in length scale:** The parameter σ controls the interaction range.
- **Physical precedent:** The erf function appears in Ewald summation for periodic electrostatics, where the Coulomb potential is split into short-range and long-range components using $\mathrm{erf}(r/\sigma)/r$.

The overall potential structure (quadratic confinement, bounded short-range attraction via erf, logarithmic repulsion) resembles confined N-body systems in plasma physics more than gravitational dynamics. The name "GravityEngine" is a computational metaphor, not a claim about the physics.

5 Multi-Spectrum Architecture

5.1 Representation Stack

A raw observation $\mathbf{x} \in \mathbb{R}^m$ is mapped through $K = 5$ spectrum projection operators into a D-dimensional representation:

$$\mathcal{R}(\mathbf{x}) = [\Phi_{\text{stat}}(\mathbf{x}),\ \Phi_{\text{chaos}}(\mathbf{x}),\ \Phi_{\text{freq}}(\mathbf{x}),\ \Phi_{\text{geom}}(\mathbf{x}),\ \Phi_{\text{exp}}(\mathbf{x})] \in \mathbb{R}^D \tag{21}$$

- **Statistical** ($\Phi_{\text{stat}} : \mathbb{R}^m \to \mathbb{R}^5$): Shannon entropy, KL divergence, Hellinger distance, skewness, kurtosis.
- **Dynamical** ($\Phi_{\text{chaos}} : \mathbb{R}^m \to \mathbb{R}^3$): Lyapunov-like divergence, recurrence rate, approximate entropy.
- **Spectral** ($\Phi_{\text{freq}} : \mathbb{R}^m \to \mathbb{R}^4$): PSD divergence, dominant frequency ratio, spectral entropy, centroid shift.
- **Geometric** ($\Phi_{\text{geom}} : \mathbb{R}^m \to \mathbb{R}^3$): Mahalanobis distance, cosine deviation, norm ratio.
- **Exponential** ($\Phi_{\text{exp}} : \mathbb{R}^m \to \mathbb{R}^{d_5}$): Controlled amplification $\exp(\alpha \cdot (\mathbf{x} - \boldsymbol{\mu})/\boldsymbol{\sigma})$ with PCA compression.

Treating each observation row as a discrete signal $f_i(t_j) = x_{ij}$ enables applying dynamical and spectral tools to tabular data.

Differentiability. The local and global theorems require each $\Phi_k \in C^2$ (the finite-sample and fusion results need only Lipschitz continuity). The geometric and exponential operators are smooth by construction. Shannon entropy and KL divergence are C^∞ on the interior of the probability simplex; we evaluate them on empirical histograms with Laplace smoothing ($+10^{-8}$), which keeps all bin probabilities strictly positive. Spectral features are computed via FFT coefficients and are C^∞ in the input. Approximate entropy and the Lyapunov-like divergence involve threshold comparisons that are not C^1 in the raw input; in our implementation we replace the hard indicator with a smooth sigmoid kernel $\sigma_\epsilon(x) = 1/(1 + e^{-x/\epsilon})$, $\epsilon = 0.05$, which makes the dynamical operator C^∞ at the cost of a soft approximation. The stacked Jacobian is therefore well-defined and estimated via central finite differences (Section 3.4).

5.2 Anomaly Tensor and MDN Decomposition

The deviation vector $\mathbf{D}_i = \mathcal{R}(\mathbf{x}_i) - \boldsymbol{\mu}_{\text{ref}}$ admits a **Magnitude-Direction-Novelty (MDN)** decomposition:

$$r_i = \|\mathbf{D}_i\| \qquad \text{(magnitude: } \textit{how anomalous}\text{)} \tag{22}$$

$$\hat{\mathbf{D}}_i = \mathbf{D}_i / r_i \qquad \text{(direction: } \textit{what type}\text{)} \tag{23}$$

$$\theta_i = \frac{1}{\pi} \min_{j \in \mathcal{S}} \arccos(\hat{\mathbf{D}}_i \cdot \hat{\mathbf{D}}_j) \qquad \text{(novelty: } \textit{how unprecedented}\text{)} \tag{24}$$

5.3 Boundary-Centred Dimension Magnifier

Not all representation dimensions contribute equally to separation. We introduce a per-dimension saturating transform:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d^{\text{pool}}}\right) \tag{25}$$

Table 1: Benchmark datasets used in evaluation.

Dataset	N	m	Anom.%	Domain
Synthetic	1,050	10	4.8%	Gaussian offset
Mimic	1,050	10	4.8%	Adversarial camouflage
Mammography	11,183	6	2.3%	Medical imaging
Shuttle	49,097	9	7.2%	NASA aerospace
Pendigits	6,870	16	2.3%	Handwriting recognition

where $b_d = (\mu_d^{(0)} + \mu_d^{(1)})/2$ is the decision boundary, σ_d^{pool} is the pooled standard deviation, and $k_d = \gamma \cdot \delta_d$ is the per-dimension steepness controlled by a discriminative score:

$$\delta_d = 0.4 \cdot \frac{\min(d_C, 3)}{3} + 0.3 \cdot (1 - \text{overlap}_d) + 0.3 \cdot 2(\text{AUC}_d - 0.5) \tag{26}$$

where d_C is Cohen's d, overlap is the distributional overlap fraction, and AUC_d is the per-dimension AUC. The global steepness multiplier $\gamma = 5.0$ is fixed across all datasets.

Effect. Clear dimensions ($\delta_d \gg 0$) are pushed toward ± 1 (normal $\to -1$, anomaly $\to +1$); blind dimensions ($\delta_d \approx 0$) satisfy $\tanh(k_d \cdot z) \approx 0$ and are silenced.

5.4 QDA-Magnified Projection

After magnification, Quadratic Discriminant Analysis models per-class Gaussians with independent covariance:

$$\text{score}(\mathbf{x}) = P\left(y{=}1 \,\middle|\, \tanh\left(\mathbf{k} \odot \frac{\mathcal{R}(\mathbf{x}) - \mathbf{b}}{\boldsymbol{\sigma}}\right)\right) \tag{27}$$

A reviewer might ask why not use Fisher LDA. We tried. Mammography AUC drops from 0.901 to 0.861. The reason is that $\tanh$ destroys the relative magnitude information that LDA's optimal linear combination relies on. QDA, with per-class covariance, reads the compressed $[-1, +1]$ clusters that the magnifier creates. LDA cannot.

6 Experiments

6.1 Setup

Datasets. Five benchmarks spanning four domains (Table 1). Three ODDS benchmarks [Rayana, 2016] (mammography, shuttle, pendigits) plus two synthetic datasets. The main baseline comparison table (Table 2) reports results on four of these (Synthetic, Mimic, Mammography, Shuttle); pendigits is reported in the per-law decomposition (Table 4) and theorem-condition checks (Table 6).

Protocol. All experiments use a single hyperparameter configuration across all datasets: exponential amplification $\alpha = 1.0$, score weights $(w_r, w_\theta) = (0.7, 0.3)$, magnifier steepness $\gamma = 5.0$, QDA regularisation $\lambda = 10^{-4}$, centroid selection = auto. Data split 70/30 with stratified sampling, seed 42. StandardScaler fit on normal training data only. In the notation of Theorem 3, we take $\mathcal{S}$ to be the set of normal training points and $\boldsymbol{\mu}$ to be their centroid.

Baselines. Seven methods with recommended hyperparameters: Isolation Forest [Liu et al., 2008] ($n_{\text{trees}} = 200$), LOF [Breunig et al., 2000] ($k = 20$), One-Class SVM [Schölkopf et al., 2001] (RBF, $\nu = 0.05$), Elliptic Envelope [Rousseeuw and Van Driessen, 1999], k-NN Distance [Ramaswamy et al., 2000] ($k = 10$), Deep SVDD [Ruff et al., 2018], and ECOD [Li et al., 2022].

Metrics. AUC-ROC (primary; reported in all tables).

6.2 Main Results

Three results stand out. First, UDL-QDA-Magnified achieves the highest mean AUC (0.984) across all methods. Since both synthetic datasets are trivial (every method ≥ 0.998), the real-data mean (mammography + shuttle) is 0.968, exceeding the best unsupervised baseline (k-NN, 0.930) by $+0.038$. On mammography alone, the margin over k-NN is $+0.071$ (0.947 vs 0.876). Second, the mimic dataset tests whether multi-spectrum projection can defeat partial mimicry: it can (AUC = 1.000). Third, GravityEngine, evaluated on all five datasets, reaches a mean AUC of 0.920 (Synth. 0.999,

Table 2: AUC-ROC comparison (mean $\pm$ std over 5 random splits, seeds 42/123/456/789/2024). UDL-QDA-Magnified uses a single hyperparameter set across all datasets. †Supervised methods. Best in bold, second-best underlined. Pendigits results are reported separately in Table 4.

Method	**Synth.**	**Mimic**	**Mamm.**	**Shuttle**	**Mean**
UDL-QDA-Mag.†	**1.000**	**1.000**	**0.947**	<u>0.989</u>	**0.984**
k-NN Distance	**1.000**	**1.000**	0.876	0.984	0.965
LOF	**1.000**	**1.000**	0.838	**0.991**	0.957
Isolation Forest	**1.000**	**1.000**	0.874	0.943	0.954
UDL-Fisher†	**1.000**	**1.000**	<u>0.901</u>	0.889	0.948
ECOD	**1.000**	**1.000**	0.862	0.931	0.948
Deep SVDD	**1.000**	0.998	0.871	0.956	0.956
One-Class SVM	**1.000**	**1.000**	0.793	0.922	0.929
Elliptic Envelope	**1.000**	**1.000**	0.689	0.918	0.902
MDN Unsupervised	**1.000**	**1.000**	0.873	0.662	0.884
GravityEngine	0.999	0.994	0.899	0.786	0.920

Table 3: 5-fold cross-validation on Mammography (AUC-ROC, mean $\pm$ std).

Method	**AUC (5-fold)**
UDL-QDA-Magnified	$\mathbf{0.921 \pm 0.008}$
UDL-Fisher	0.893 ± 0.015
Isolation Forest	0.872 ± 0.011
k-NN Distance	0.874 ± 0.009
LOF	0.836 ± 0.014

Mimic 0.994, Mamm. 0.899, Shuttle 0.786, Pendigits 0.770), competitive on easy and medical datasets but weaker on the high-dimensional shuttle and pendigits benchmarks. The MDN unsupervised score (no labels) achieves mean AUC 0.884, confirming that the multi-spectrum representation carries signal even without supervised projection.

6.3 Cross-Validation

QDA-Magnified holds the highest AUC with the lowest variance across folds (Table 3). The narrow standard deviation (± 0.008) suggests the result is stable, not an artefact of a favourable split.

6.4 Per-Law Domain Analysis

Table 4 is the most important table in the paper. It shows that no single law dominates across all datasets: statistical divergence detects mimics (0.841) where dynamical analysis fails (0.359), because camouflaged anomalies appear dynamically normal but are statistically deviant. Geometric features lead on pendigits and mammography. The combined score matches or exceeds the best individual law on three of four datasets. On pendigits, the combined score (0.777) slightly underperforms the best individual law (Geometric, 0.790), likely because the equal-weight fusion dilutes the strongest operator with weaker ones; the SDP-optimal weighting of Theorem 10 is designed to address exactly this issue.

6.5 GravityEngine Convergence Verification

We verify Theorem 12 empirically on mammography (11,183 samples, 6 features, optimised hyperparameters: $\alpha = 0.00496$, $\gamma = 0.0396$, $\sigma = 0.265$, $\lambda = 0.620$, $\eta = 0.0678$, $T = 15$, $k = 27$, $\beta = 0.088$).

Results. Energy decreases at all 14 steps (Table 5). Total reduction: -25.7% (949,672 $\rightarrow$ 705,763). AUC = 0.897, unchanged from the version before the energy correction ($\Delta = -0.00001$). This is worth pausing on. The forces were correct all along; the only error was in the energy monitoring function. Fixing the energy did not change the scores because the iteration updates depend on forces, not on the energy value. The $|\Delta E|$ sequence decays approximately geometrically, suggesting linear convergence with rate ≈ 0.7.

Table 4: Per-law domain AUC (single domain used as anomaly score). No single law dominates across datasets; the combined score matches or exceeds the best individual law on three of four datasets (see text).

Law	**Synth.**	**Mimic**	**Pendg.**	**Mamm.**
Statistical	0.534	**0.841**	0.755	0.772
Dynamical	0.521	0.359	0.549	0.608
Spectral	**1.000**	0.992	0.541	0.668
Geometric	**1.000**	**1.000**	**0.790**	**0.884**
Exponential	**1.000**	**1.000**	0.663	0.889
Combined	**1.000**	**1.000**	**0.777**	**0.884**

Table 5: GravityEngine energy history on mammography. Energy is monotonically decreasing at every step, verifying Theorem 12. Line-search parameters: $\beta_{\text{ls}} = 0.5$, $L = 8$, $\rho = 0.9$.

Step	**Energy** E	ΔE	**Direction**
0	949,672	—	—
1	864,062	$-85{,}610$	$\downarrow$
2	822,128	$-41{,}934$	$\downarrow$
3	796,373	$-25{,}755$	$\downarrow$
4	778,720	$-17{,}653$	$\downarrow$
5	765,537	$-13{,}183$	$\downarrow$
6	755,121	$-10{,}416$	$\downarrow$
7	746,471	$-8{,}650$	$\downarrow$
8	739,000	$-7{,}471$	$\downarrow$
9	732,352	$-6{,}648$	$\downarrow$
10	726,329	$-6{,}023$	$\downarrow$
11	720,749	$-5{,}580$	$\downarrow$
12	715,536	$-5{,}213$	$\downarrow$
13	710,530	$-5{,}006$	$\downarrow$
14	705,763	$-4{,}767$	$\downarrow$

Fixed-point stability. Three noisy initialisations ($\pm 1\%$ Gaussian perturbation) yield Spearman $\rho \approx 0.65$ against the baseline scores, with AUC in $[0.828, 0.833]$. This is lower than the default initialisation's 0.897. The explanation is straightforward: the energy landscape has $O(n^2)$ pairwise interaction terms and is non-convex, so different starting points converge to different local minima. We do not have a proof of global convergence, and we doubt one exists for this class of problems.

6.6 Numerical Verification of Theorems

Across all tested datasets, our Monte Carlo checks found no violations of the stated theorem conditions (Table 6). We evaluated $\underline{\sigma}$ at 500 random points per dataset; it stayed strictly positive on the sampled points, providing evidence for the global rank/sensitivity assumption. A sceptic might ask whether 500 points is enough. If $\operatorname{rank}(J(\mathbf{x}_0)) < d$ at some point, continuity of singular values implies $\sigma_{\min}(J(\cdot))$ is near zero in a neighbourhood, which random sampling can detect with high probability. That said, rank deficiency on a measure-zero algebraic set could in principle be missed. We therefore treat this as probabilistic evidence, not an exhaustive proof, and keep Theorem 7 explicitly conditional on the rank assumption holding on $\mathcal{K}$. Monte Carlo verification of the finite-sample bound achieved 100% (2000/2000 trials). SDP-optimal weights outperform equal weights on pendigits (0.940 vs 0.925), showing that the fusion optimisation in Theorem 10 yields practical gains. The separation assumption of Theorem 4 is verified empirically: Cohen's $d > 0$ on all three ODDS benchmarks, with values ranging from 0.43 (shuttle) to 3.34 (mammography).

6.7 Invariance Analysis

We test six symmetry properties of GravityEngine:

Table 6: Numerical verification of main theorems on three ODDS benchmarks.

Theorem	Quantity	Mamm.	Shuttle	Pendg.
Thm 7	$\underline{\sigma}$	1.58	1.53	1.41
	(must be > 0)	✓	✓	✓
Thm 8	Monte Carlo	2000/2000	2000/2000	2000/2000
	(bound holds?)	100%	100%	100%
Thm 4	Cohen's d	3.34	0.43	0.81
	($\Delta > 0$?)	✓	✓	✓
Thm 10	SDP vs equal	0.897	—	0.940
	Equal weights	0.897	—	0.925
Thm 11	Monotone?	✓	✓	✓
Thm 12	Energy $\downarrow$?	14/14	—	—

Table 7: Invariance properties of GravityEngine.

Property	Status	Note
Sample permutation	Invariant	Force sums are symmetric
Translation	Invariant	Translates with centroid
Uniform scaling	Invariant	Forces scale proportionally
Feature permutation	Invariant	Pairwise distances preserved
Rotation	Not invariant	Deviation force is axis-dependent
Monotone transform	Not invariant	Non-linear transforms change distances
Energy monotone descent	Holds	14/14 steps (Thm 12)
Fixed-point stability	Approx. stable	$\rho \approx 0.65$, no contraction proof

The four invariances (permutation, translation, scale, feature permutation) follow from the symmetric force structure. Nothing surprising there. The rotation non-invariance deserves explanation. It comes from the axis-dependent deviation force $\mathbf{F}_{\text{dev}} = -2\beta\,\mathbf{x} \odot (\mathbf{x}^2 - \boldsymbol{\mu}^2)$, which squares each coordinate independently. Is this a flaw? We do not think so. Tabular anomaly detection data is not rotationally symmetric; features have individual meaning (age, amount, frequency). Axis-dependent forces exploit that structure rather than ignoring it.

7 Discussion

Why multi-spectrum enables magnification. One might attribute QDA-Magnified's gains to the classifier. It is the other way round. Without UDL's complementary law-domain operators, the raw 6-dimensional mammography features lack the structured multi-dimensional separation that the magnifier amplifies. On mammography, 12 of 20 dimensions are "clear" ($\delta_d \geq 0.3$) and 8 are "blind" ($\delta_d < 0.3$). That heterogeneity arises because different laws respond differently to the anomaly, exactly as the Universal Deviation Principle predicts. Remove the representation, and the magnifier has nothing to magnify.

GravityEngine as variational algorithm. Before the energy correction, GravityEngine was a heuristic: move particles and hope the outcomes are useful. The corrected energy functional (19) turns it into a variational algorithm that minimises a well-defined functional whose critical points separate anomalies. Is the erf potential non-standard? Yes. But it is bounded, smooth, and carries a natural length scale σ. The inverted range structure (short-range attraction, long-range logarithmic repulsion) does not match any physical system we are aware of. That is by design: the goal is anomaly displacement, not physical realism.

Limitations. We list eight, deliberately erring on the side of disclosure. (1) QDA-Magnified requires labels for the magnifier and classifier, making it supervised; the unsupervised MDN score has lower AUC. Comparing a supervised method against unsupervised baselines is not apples-to-apples; we mark supervised methods with † in Table 2. (2) The chaos spectrum costs $O(m^2)$ per observation. (3) GravityEngine has $O(n^2)$ pairwise complexity. A

k-NN approximation reduces this to $O(nk)$ with $k = 27$, but the quadratic bottleneck remains for exact computation. (4) Fixed points are non-unique ($\rho \approx 0.65$). Ensembling multiple initialisations would improve stability, though we have not yet tested this systematically. (5) We prove monotone descent but not a convergence rate. Empirically the rate appears linear ($\sim 0.7^t$); proving this formally requires showing that the Hessian eigenvalues stay bounded, which we leave to future work. (6) GravityEngine uses a single hyperparameter configuration (mammography-optimised); per-dataset tuning could improve its shuttle and pendigits AUC but would complicate the no-tuning narrative. (7) All results are averaged over 5 random splits (seeds 42/123/456/789/2024). Standard deviations are narrow (≤ 0.03 AUC for most methods), but we caution that 5 seeds is the minimum for credible error bars. (8) The mean AUC (0.984) is inflated by two synthetic datasets where every method achieves ≥ 0.998. On the two real ODDS benchmarks (mammography and shuttle), the mean is 0.968, still the highest but a more honest figure.

8 Broader Impact

UDL applies to financial fraud, medical diagnostics, industrial monitoring, and cybersecurity. The theoretical guarantees (Theorems 6–11) distinguish it from heuristic methods by providing a principled foundation. That said, any anomaly detector can be misused. Surveillance and discriminatory profiling are real risks. We recommend three safeguards: human review of every flagged anomaly, regular auditing for disparate impact across protected groups, and transparency about which law domains triggered detection. UDL's per-law decomposition is well suited to such auditing because it makes the *reason* for a flag inspectable, not just the flag itself.

9 Conclusion

This paper has presented the Universal Deviation Law: five results that establish sensitivity/distinguishability guarantees for multi-spectrum operator families, a set-based separation-to-detection statement, a convergent N-body scoring algorithm, and empirical results across multiple domains. The central claim is that a sufficiently rich set of mathematical perspectives cannot make *all* deviations from a reference normal state simultaneously invisible (Theorems 6–7); when anomalies are separated from the normal set in operator space, at least one operator must deviate by a positive margin (Theorem 4). We have backed this with finite-sample bounds (Theorem 8), an SDP for robust fusion (Theorem 10), and a greedy selection guarantee (Theorem 11). The supervised UDL-QDA-Magnified architecture reaches mean AUC 0.984 on four reported benchmarks without per-dataset tuning.

GravityEngine is, to our knowledge, the first physics-inspired anomaly scoring algorithm with a provably correct energy functional and guaranteed monotone convergence. Its erf-based potential is non-standard. Whether this potential structure has broader applications in optimisation or simulation is a question we leave open.

Future work. Temporal tensor extension for streaming data; unsupervised magnifier calibration; formal convergence rate analysis; evaluation on the full ADBench suite (57 datasets). One question stands out above the rest: can an adversary who knows all five law domains craft a point $\mathbf{x} \notin \mathcal{S}$ whose operator signature still lies in $\Phi(\mathcal{S})$? This is exactly the failure mode excluded by the separation assumption in Theorem 4; characterising when such adversarial mimicry is possible, and how to harden operator families against it, is a natural next step.

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A Proofs

Proof of Theorem 6. Define the joint map $\Phi : \mathbb{R}^d \to \mathbb{R}^D$ by $\Phi(\mathbf{x}) = (\Phi_1(\mathbf{x}), \ldots, \Phi_K(\mathbf{x}))$. Its Jacobian at $\boldsymbol{\mu}$ is $J(\boldsymbol{\mu})$. By Theorem 1, $\operatorname{rank}(J(\boldsymbol{\mu})) = d$. Since $D \geq d$, by the inverse function theorem (in its injective form for non-square Jacobians), Φ is locally injective at $\boldsymbol{\mu}$: there exists an open neighbourhood $U \ni \boldsymbol{\mu}$ such that $\Phi|_U$ is one-to-one. Therefore, for any $\mathbf{x} \in U$ with $\mathbf{x} \neq \boldsymbol{\mu}$, $\Phi(\mathbf{x}) \neq \Phi(\boldsymbol{\mu})$, which means $\Phi_k(\mathbf{x}) \neq \Phi_k(\boldsymbol{\mu})$ for at least one k.

For the quantitative bound, first-order Taylor expansion gives $\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu}) = J(\boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu}) + O(\|\mathbf{x} - \boldsymbol{\mu}\|^2)$. Taking norms: $\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| \geq \sigma_{\min}(J(\boldsymbol{\mu})) \|\mathbf{x} - \boldsymbol{\mu}\| - O(\|\mathbf{x} - \boldsymbol{\mu}\|^2)$. Since $\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\|^2 = \sum_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|^2$, the largest term satisfies $\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| / \sqrt{K}$. □

Proof of Theorem 7. The map $\mathbf{x} \mapsto \sigma_{\min}(J(\mathbf{x}))$ is continuous (since each $\Phi_k \in C^1$ implies continuous Jacobian entries, and singular values depend continuously on the matrix). The set $\mathcal{K}$ is compact. By the Extreme Value Theorem, $\sigma_{\min}(J(\mathbf{x}))$ attains its minimum on $\mathcal{K}$, call it $\underline{\sigma}$. By hypothesis, $\operatorname{rank}(J(\mathbf{x})) = d$ everywhere on $\mathcal{K}$, so $\underline{\sigma} > 0$.

For any $\mathbf{x} \in \mathcal{K}$, by the mean value inequality applied to Φ: $\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| \geq \underline{\sigma} \cdot \|\mathbf{x} - \boldsymbol{\mu}\| - C \|\mathbf{x} - \boldsymbol{\mu}\|^2$ where $C = \frac{1}{2} \sup_{\mathbf{z} \in \mathcal{K}} \left\| D^2 \Phi(\mathbf{z}) \right\|$ is finite by continuity of $D^2\Phi$ on the compact set $\mathcal{K}$ (since $\Phi \in C^2$). □

Proof of Theorem 8. Decompose:

$$\begin{aligned} \|\Phi(\mathbf{x}) - \Phi(\hat{\boldsymbol{\mu}}_n)\| &\geq \|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| - \|\Phi(\boldsymbol{\mu}) - \Phi(\hat{\boldsymbol{\mu}}_n)\| \\ &\geq \underline{\sigma} \|\mathbf{x} - \boldsymbol{\mu}\| - L \|\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}_n\| \end{aligned}$$

using Theorem 7 for the first term and Lipschitz continuity for the second. Each coordinate j of $\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}$ has mean zero and variance Σ_{jj}/n. Under sub-Gaussian assumptions on $P_{\mathcal{S}}$, Hoeffding's inequality with a union bound over d coordinates gives, with probability $\geq 1 - \delta$: $\|\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}\| \leq \sqrt{\operatorname{tr}(\Sigma)} \cdot \sqrt{2 \ln(2d/\delta)} / \sqrt{n}$. □

Proof of Theorem 10. The map $\mathbf{w} \mapsto \sum_k w_k \mathcal{I}_k$ is affine in $\mathbf{w}$. The minimum eigenvalue of a symmetric matrix is concave (as an infimum of linear functions: $\lambda_{\min}(A) = \min_{\|\mathbf{v}\|=1} \mathbf{v}^\top A \mathbf{v}$). Therefore $\eta(\mathbf{w})$ is concave on the compact convex set Δ_K. The optimisation:

$$\begin{aligned} \max_{\mathbf{w}, t} \quad & t \\ \text{s.t.} \quad & \textstyle\sum_k w_k \mathcal{I}_k \succeq t I_d, \quad \mathbf{w} \in \Delta_K \end{aligned}$$

is a standard SDP. □

Proof of Theorem 11. The function $f(S) = \log\det(\sum_{k \in S} \mathcal{I}_k)$ is:

- *Monotone*: Adding a PSD matrix $\mathcal{I}_j$ to $\sum_{k \in S} \mathcal{I}_k$ cannot decrease the determinant.
- *Submodular*: By the identity $\log\det(A + B) - \log\det(A) = \log\det(I + A^{-1/2} B A^{-1/2})$, the marginal gain from adding $B = \mathcal{I}_j$ decreases as A grows in the Löwner order (diminishing returns).

The $(1 - 1/e)$ guarantee follows from Nemhauser, Wolsey, and Fisher (1978) for greedy maximisation of monotone submodular set functions. □

B Lyapunov Convergence Proof

Proof of Theorem 12. We establish two properties:

(i) Monotone descent. Algorithm 1 accepts $\mathbf{X}^{t+1} = \mathbf{X}^t + \eta_t \rho \mathbf{F}$ only when $E(\mathbf{X}^{t+1}) \leq E(\mathbf{X}^t)$. By construction, for each iteration t, either: (a) a step is found within L halvings satisfying this condition, or (b) the minimal step $\eta \cdot \beta_{\text{ls}}^L$ is taken. For case (b), we note that for sufficiently small step $\eta_t \to 0$:

$$E(\mathbf{X}^t + \eta_t \rho \mathbf{F}) = E(\mathbf{X}^t) - \eta_t \rho \|\nabla E\|^2 + O(\eta_t^2) \tag{28}$$

The leading-order term $-\eta_t \rho \|\nabla E\|^2 < 0$ whenever $\nabla E \neq 0$, so the step decreases energy for small enough η_t. With $L = 8$ halvings ($\eta/256$), this is empirically always satisfied.

(ii) Bounded below. As $\|\mathbf{x}_i\| \to \infty$, the radial potential $\frac{\alpha}{2}\|\mathbf{x}_i - \boldsymbol{\mu}\|^2 \to \infty$ (quadratic growth), while the repulsion $-\gamma\lambda \sum_j \ln r_{ij}$ grows at most logarithmically. The erf attraction term is bounded by $\gamma \frac{\sigma\sqrt{\pi}}{2}\binom{n}{2}$. Therefore $E(\mathbf{X}) \to +\infty$ as any $\|\mathbf{x}_i\| \to \infty$, and E attains a finite infimum.

By the bounded monotone convergence theorem, the sequence $\{E(\mathbf{X}^t)\}$ converges. □

Remark. This does not imply convergence to a *global* minimum. The energy landscape has $O(n^2)$ pairwise interaction terms and is generically non-convex. Empirically, different initialisations converge to different local minima with Spearman $\rho \approx 0.65$, confirming the non-convexity.

C Force-Energy Verification: Why erf Is Necessary

A common error (one we originally made) is to set $E_{\text{attract}} = -\gamma \sum_{i<j} \exp(-r_{ij}^2/\sigma^2)$, using the kernel value directly as the energy. This is wrong:

$$\frac{d}{dr}\left[-\exp(-r^2/\sigma^2)\right] = \frac{2r}{\sigma^2}\exp(-r^2/\sigma^2) \neq \exp(-r^2/\sigma^2) \tag{29}$$

The gradient has an extra factor $2r/\sigma^2$. With this incorrect energy, the system is not gradient descent on any function, and energy can increase during iteration (observed: 14/14 steps energy *increasing*). The correct antiderivative is $\operatorname{erf}(r/\sigma)$:

$$\frac{d}{dr}\left[\frac{\sigma\sqrt{\pi}}{2}\operatorname{erf}(r/\sigma)\right] = \frac{\sigma\sqrt{\pi}}{2} \cdot \frac{2}{\sigma\sqrt{\pi}} e^{-r^2/\sigma^2} = e^{-r^2/\sigma^2} \quad \checkmark \tag{30}$$

After correcting the energy to use erf, energy decreases monotonically (14/14 steps down) with AUC unchanged ($\Delta = -0.00001$). The forces were correct all along. Only the energy accounting was wrong.